picBunch: a free-space FFT convolution for the initial field of a uniformly moving particle bunch, and its measured cost and accuracy

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Run `pr24-20261102`

1 Introduction

When a particle-in-cell (PIC) simulation starts, the bunch of particles must carry an initial electromagnetic field, or the first push already acts on the wrong force. The field of a moving bunch in free space is known in closed form for each point charge, but a simulation reads that field on a staggered mesh with thousands of observation points, so it is computed by a fast sum over the source charges rather than by evaluating the closed form everywhere. We do that sum as a TSC deposit onto a zero-padded grid, a sampled exact TSC-convolved kernel, and a free-space FFT convolution, with staggered Yee read-out. This paper gives the exact discrete recipe a re-implementer needs, and it measures the cost and the accuracy of `picBunch v0.5.0` (commit `7b4f1766`, tag `3e083fef`) on a single V100, naming the reference for every error statement. The release is driven by a single TOML configuration (input file, destination, config; Section 3.1), and it carries the capabilities the measurements below exercise: an opt-in exact near shell, MPI chunked output for global domains larger than the compute box, a tolerance-derived far-tail margin for zero fill, and a float32 checkpoint that restarts under a stock PIConGPU.

The established workaround avoids computing the initial field altogether: one starts from a field-free state and marches the bunch forward, letting the fields build up in the first few steps. The present-time field of a uniformly moving bunch has no transient, so this history has no physical basis. We measure how well the discrete method reproduces that field and how much it costs, and we state the conditions under which it fails (Section 5).

2 Field definition

The target quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p, and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation point $\mathbf{x}_{\mathrm{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\mathrm{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\mathrm{point},c}\big(\mathbf{x}_{\mathrm{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 3) and K_{point} is the per-unit-charge field of one uniformly moving charge. Because the integral is linear in the source charge, the sum over particles is exact; the whole design is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge [extraction worker, 2026]. For constant velocity the field points to the instantaneous position. In the lab frame, with $\mathbf{R} = \mathbf{x}_{\mathrm{obs}} - \mathbf{x}_{\mathrm{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \mathrm{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2\, \mathrm{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \mathrm{invg}^2\, \mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class; the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The reference normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta}\times E_n$ [extraction worker, 2026]. The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m and $c = 299792458$ m/s. The h^3 factor (Section 3) is the Riemann volume of the sampled continuum integral [extraction worker, 2026].

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis [extraction worker, 2026]:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Method

3.1 The operator interface

`v0.5.0` (commit `7b4f1766`, tag `3e083fef`) is driven by `picBunch -input bunch.h5 -destination <checkpoint> -config params.toml`, with every unit and conversion parameter in the TOML file [worker, 2026a]. The `[picongpu]` section carries `unit_length`, `unit_e`, `unit_b`, `dt`, `step`, `precision` and `species`; `[domain]` carries `cells`, `cell_size` and `origin`; `[placement]` carries `offset_unit` and the `left`/`front`/`top` fractions; `[accuracy]` carries the `tolerance` and the near-shell settings. A missing conversion key is a clear failure, not a silent default. The input bunch is centred by its charge-weighted centroid and placed at the requested offset from the domain origin; a placement that leaves the bunch outside the domain is rejected, never clipped.

The default `precision` is uniform `float32` for every field and particle record, and such a checkpoint restarts under a stock single-precision PIConGPU; `float64` is a uniform opt-in that records a warning attribute because stock PIConGPU cannot load double records [worker, 2026b]. The `charge` and `mass` records are written rather than omitted, and the species is a single particle patch, `patch` = `"bbox"` or `"domain"` [worker, 2026a].

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, and a free-space FFT convolution. Let N be the bunch-box nodes per axis, $M = \mathrm{pad}\cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at ih **cell**), and $\mathbf{cell} = (dx, dy, dz)$. All precision-critical work is in double [worker, 2026c, extraction worker, 2026].

3.2 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h [extraction worker, 2026]. The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\mathrm{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd` [extraction worker, 2026].

3.3 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components

c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\,\mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.5). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide [worker, 2026d]. Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT [extraction worker, 2026].

3.4 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution. The vendor plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transformations [extraction worker, 2026]. The unnormalized round trip is scaled by M^3; the code divides by h^3M^3 in the read-out [extraction worker, 2026].

Padding. The grid must be zero-padded to pad ≥ 2. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box [worker, 2026e]. At pad $= 2$ the result agrees with pad $= 3$ to 8.4×10^{-16} and with a direct non-cyclic sum to 2.8×10^{-17} [worker, 2026e,c]. At pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$) [worker, 2026e]. The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error [worker, 2026e].

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real [worker, 2026e]. The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components [worker, 2026e].

The h^3 factor. The sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units); the read-out therefore applies the factor $h_x h_y h_z/(M_x M_y M_z)$ [extraction worker, 2026].

3.5 Read-out

The six field components live at the staggered Yee positions, in cell units [extraction worker, 2026]:

$$\begin{array}{llllll} E_x & (0.5,0,0) & E_y & (0,0.5,0) & E_z & (0,0,0.5) \\ B_x & (0,0.5,0.5) & B_y & (0.5,0,0.5) & B_z & (0.5,0.5,0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed [extraction worker, 2026].

3.6 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units [worker, 2026d]. At a fixed distance in cells the error is therefore independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1, 1/2, 1/4, 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4 [worker, 2026d]. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The only knob is r/h: reaching 10^{-3} perpendicular at $\gamma = 10$ for $r = 8$ cells needs $h \approx 1/29$, 2.4×10^4 times the M^3 nodes of the $h = 1$ run [worker, 2026d, Orchestrator, 2026a].

3.7 The exact near shell

The default mesh is the whole field. `v0.5.0` carries an opt-in exact near shell (`[accuracy]` in the TOML config, Section 3.1), with a radius and a tile [worker, 2026f,g]. It computes

$$F = F_{\text{mesh}}(\text{all}) - F_{\text{mesh}}(\text{near}) + F_{\text{exact}}(\text{near}), \tag{9}$$

Table 1: Far-tail switch validation, 20-cell 4096-electron bunch, $R = 16.85$ cells, target 45.72 V/m (10^{-3} of the peak 45722.5 V/m). Worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum over 13 directions [worker, 2026g]. The switch at $6R$ is the one shipped: the error is 1.87×10^{-3} of the target.

switch radius	worst err (V/m)	/ target
$2R$ (33.7 cells)	4.80×10^{1}	1.05 (fails)
$4R$ (67.4 cells)	8.77×10^{-1}	1.92×10^{-2}
$6R$ (101.1 cells)	8.53×10^{-2}	1.87×10^{-3}
$8R$ (134.8 cells)	1.66×10^{-2}	3.63×10^{-4}

where the subtraction uses a per-tile local zero-padded FFT of the near sources (padding 2) and the exact term is a vendored FP64 TSC-convolution reference evaluated per (source, node, component) [worker, 2026f]. The default remains `none`; the radius is in mesh nodes of the compute grid, and the envelope it must cover depends on h (Section 6.5).

The assembly is exact. Single-source tests at a tile centre and a tile boundary give an assembly residual of 2×10^{-12} to 1.5×10^{-11} SI (4×10^{-16} relative), flat with distance. A local FFT of the near set against the global FFT agrees at 10^{-15} for margins 0, 2, 4 and a 100-particle cloud, and boundary accounting shows the same near set feeding the exact sum and the subtracted mesh (field identity 3.6×10^{-13}). A direct reproduction with 100 particles at radius 3 gives an assembled field exact to 4.6×10^{-13} over the whole grid, while the plain mesh on the same plane is wrong by 5.6×10^{-1}. A plane error of 1.36×10^{-1} is a mesh-outside-the-shell number, not an assembly error; the tile subtraction is not the limiting term [worker, 2026h].

3.8 MPI chunked output

`v0.5.0` writes a global output domain larger than the compute box [worker, 2026i,g]. The written extent is set by `-global-grid` and `-global-origin`. The global array is never allocated: each rank owns a contiguous z-slab split into tiles of about 128 MiB, zero-filled, then filled with the compute-box overlap, the rest exact 0.0. All ranks open one openPMD series with `MPI_COMM_WORLD` and write disjoint `storeChunk` regions with collective flushes, so the output is a single dataset, not per-rank files. The FFT is not domain-decomposed: only rank 0 computes the padded bunch box, and the write is decomposed. A `-deterministic-deposit` flag gives a serial host deposit for bitwise comparisons [worker, 2026i].

3.9 The far-tail margin

For a global domain that extends beyond the compute box, the zero fill must not start until the field is below tolerance. `v0.5.0` places that boundary at the smallest centre distance d where

$$E_{\text{tail}}(d) = \gamma_{\max}\, k \left(\frac{|Q|}{d^2} + \frac{2|\mathbf{p}|}{d^3} + \frac{3.5\,\|D\|_2}{d^4} \right) \le \text{tolerance} \cdot E_{\text{peak}}, \tag{10}$$

with Q the total charge, $\mathbf{p}$ the dipole, $D_{ij} = \sum q\,(3x_i x_j - |\mathbf{x}|^2\delta_{ij})$ the traceless quadrupole from the actual particle positions, $\|D\|_2$ its largest absolute eigenvalue, and $\gamma_{\max} = \max\sqrt{1+|\mathbf{u}|^2}$ [worker, 2026g]. Beyond a measured switch radius the writer instead evaluates the multipole field directly (`FarTail.hpp`), so the shipped file has no interior zero fill [worker, 2026g]. The margin is reported at run time as a bound in V/m.

For the documented 20-cell, 4096-electron bunch (source radius $R = 16.85$ cells, peak 45722.5 V/m, target 45.72 V/m), the quadrupole margin is 350 cells, which exceeds the 2^{28} padded-node cap, so the code falls back to the measured far-tail switch. Table 1 gives the switch error: the worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum in 13 directions [worker, 2026g].

A shipped 512^3 output sampled outside the compute box confirms the switch in practice: 36 nodes including corners, worst $|\mathbf{E}_{\text{file}} - \mathbf{E}_{\text{direct}}| = 2.8 \times 10^{-3}$ V/m against the 45.7 V/m target (6.1×10^{-5} of it), and every sample nonzero, i.e. no interior zero fill [worker, 2026g].

Table 2: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0 [worker, 2026c]. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

The in-support field. Inside the TSC support the singular part of the exact term is analytic: the eight `asinh` face integrals of the kernel's divergence, $O(1)$, verified against a converged trapezoid reference (-2.6333 versus -2.6348); the smooth remainder is integrated by octant split. For a moving bunch the power-transform quadrature is the validated path, device against reference 1.000000–1.000040 at $\gamma = 1/2/10/100$ (exact to the six digits shown at $\gamma = 1$ and 2), converging to $1.5{\times}10^{-6}$ at box order 32 [worker, 2026j, pr24w53 ref, 2026]; there is no validated closed form for the moving case.

4 Measured scaling

All timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0 [worker, 2026c]. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double [worker, 2026c]. Each configuration ran three timed repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max [worker, 2026c]. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run [worker, 2026c]. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 3, a different measurement path.

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold [worker, 2026c,k]. This confirms the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff.

The scaling is in two regimes, with the crossover set by the box, not the particle count [worker, 2026c]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms) [worker, 2026c]. At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× because the deposit stays constant [worker, 2026c]. The per-h totals are 64.76, 69.27, and 151.51 ms for $h = 1$, 1/2, 1/4 (padded array $M = 60/100/180$, 2×10^6 particles, 20-cell box) [worker, 2026c].

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB [worker, 2026c]. Memory scales with the padded bunch box, not the global field.

Table 3 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 → 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 2; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer [worker, 2026l,c].

4.1 MPI chunked output for global domains

The v0.5.0 chunked writer (Section 3.8) was measured at 512^3, tolerance 10^{-3}, a 20-cell 4096-electron bunch, `-deterministic-deposit`, on `rosi_v100` [worker, 2026g]. The quadrupole margin for this bunch exceeds the 2^{28} padded-node cap, so the code falls back to the measured $6R$ far-tail switch: compute

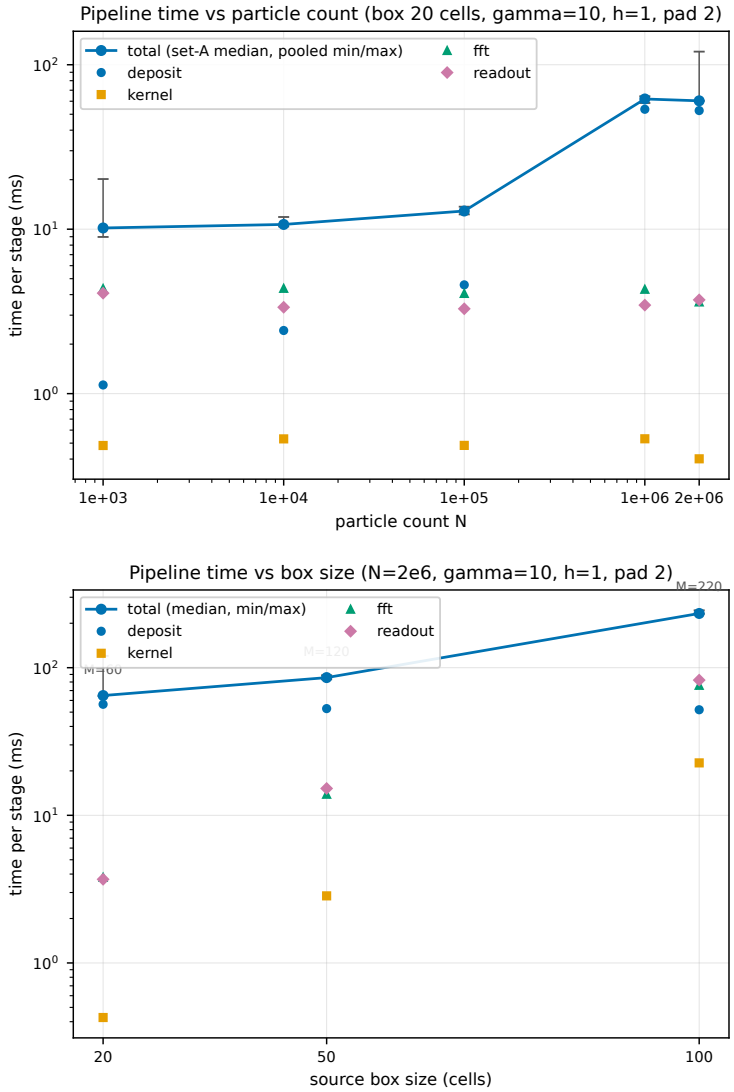


Figure 1: Pipeline time on `rosi_v100` (V100-SXM2-32GB, idle), CUDA 12.6, Release, $\gamma = 10$, $h = 1$, padding 2, double, $N = 2 \times 10^6$ for the box sweep [worker, 2026c]. Top: total time vs particle count, $N \in \{10^3, \ldots, 2 \cdot 10^6\}$, 20-cell box; grey bars span the pooled min/max over the four $2 \cdot 10^6$ repeats. Bottom: total vs source-box size 20/50/100 cells (padded array $M = 60/120/220$). Blue is the total; the other series are per-stage medians, side by side, not stacked. Log-log.

box 212^3, output 6.44 GB, six double components. Table 4 gives the wall and writer times for one and two ranks.

Table 4 shows the output is a single collective dataset, not per-rank files: the six datasets from the one-rank and two-rank runs are bitwise identical, 6 of 6 exact, $\max_{\text{abs diff}} = 0.0$, metadata equal [worker, 2026g]. The writer time is flat in rank count because each rank writes its own z-slab of the same total bytes; the wall time drops with two ranks because the slab writes overlap.

5 Limitations

The mesh is a far-field method. Its limits are set by r/h and γ, and they are measured, not assumed. The mesh alone does not reach 10^{-3} where the slices cut the bunch core (Section 6.5); that is the single-mesh limitation the slices display. The opt-in near shell (Section 3.7) assembles exactly to roundoff, so at the measured radii the residual is the mesh outside the shell, not the tile subtraction. The perpendicular direction at $\gamma = 10$ does not converge at the natural cell size: 1.5×10^{-2} at $d = 48$, aspect 1, against the point field, of which the TSC smoothing alone is 1.48×10^{-2} (Table 5) [worker, 2026d, Orchestrator,

Table 3: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, shipped CLI `-tolerance` 10^{-3} [worker, 2026l]. A different measurement path from the compute-only Table 2: the totals include the writer (the dominant added term); the per-stage split is deposit/kernel/FFT/readout/writer. The $N = 10^6$ total is 47.15 ms here versus 61.92 ms in the compute-only campaign.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

Table 4: MPI chunked output, 512^3 global domain, tolerance 10^{-3}, 20-cell 4096-electron bunch, `-deterministic-deposit`, `rosi_v100` [worker, 2026g]. Compute box 212^3 (the $6R$ far-tail switch, not the quadrupole margin, which exceeds the 2^{28} cap); output 6.44 GB. All six datasets bitwise identical across rank counts; metadata equal; $n_{\text{diff}} = 0$.

ranks	wall (s)	writer (s)
1	54.3	39.8
2	43.2	39.8

2026b]. At $\gamma = 100$ the perpendicular error is 3.75×10^{-1} at $d = 48$ and does not converge as d grows [worker, 2026d, Orchestrator, 2026c]. At aspect 4, $\gamma = 1$ already fails the gate at $d = 48$ (2.4×10^{-3}) [worker, 2026d]. The measured near-shell cost ladder does not fit the 2×10^6 / 10^{-3} / 300 s envelope at production size (Section 6.7). The detailed discussion is in Section 8.

6 Accuracy

Each error figure names its reference (M mesh, T continuum, P point). The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

6.1 The reference and the two metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map [pr24w53 ref, 2026]. Its quadrature error is below the 3×10^{-4} gate in every configuration measured here [worker, 2026d].

Three metrics are reported and are not interchangeable, and each claim in this section names which one supports it [worker, 2026m].

Floored. $|\text{ref} - \text{mesh}| / \max(|\text{ref}|, |\text{mesh}|)$, over points where $|\text{ref}| \geq 10^{-3}$ of the local (six-component) field peak. This is the relative per-component error of record for the electric field.

Unfloored. The same ratio at every point. It is the only one that reaches the magnetic field, because the floor excludes it (below).

Absolute. $|\text{ref} - \text{mesh}| / \max_{\text{point}} |\text{ref}_c|$, the peak-normalized error; it supports whole-field statements.

The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, \text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the $+1$ denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate [worker, 2026l].

The floor hides the magnetic field. For $\gamma = 10$ every B point is excluded by the floor: $|\mathbf{B}|$ is about 10^{-10} of the E-dominated local field, far below the 10^{-3} threshold, so the floored metric reports no magnetic accuracy at all. The unfloored B relative error is 9.7×10^{-4} at $d = 32$ (single particle, aspect 1), against a B peak that the floored column never sees [worker, 2026m]. The floored column of Table 6 is therefore electric-only at $\gamma = 10$.

Table 5: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; single-component worst: $\gamma = 1$, aspect 4, $M-T$ 2.65×10^{-5}; $\gamma = 10$, aspect 1, 1.4×10^{-3}) [worker, 2026d]. $M-T$: mesh vs TSC-convolved continuum; $M-P$: mesh vs point field; $T-P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1 the $M-P$ vs $M-T$ gap is the smoothing (L2 ratio 58×), not the mesh.

γ	aspect	$M-T$	$M-P$	$T-P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

Table 6: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the shipped CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; rosi_v100, `indfield.py` order 12 refine 3 [worker, 2026l]. "s/c" = single / cloud. $\gamma = 10$ does not cross 3×10^{-4} within 48 cells (6.5×10^{-3} at $d = 32$, 1.8×10^{-3} at $d = 48$); $\gamma = 1$ crosses at $d = 12$ (aspect 1) and $d = 32$ (aspect 4).

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

6.2 Reference-separated error at 48 cells

Table 5 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M-P = 1.5 \times 10^{-2}$) is about 58× the mesh against its own target ($M-T = 2.6 \times 10^{-4}$, an L2 ratio); the difference is the TSC smoothing, which $M-P$ inherits and $M-T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M-T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ [worker, 2026d].

6.3 The independent validation

A second, independent pass measured the shipped CLI end to end: `picBunch` at `bfaa1e26`, rosi_v100, `indfield.py` order 12 refine 3, $d = 2$ to 32 cells, $\gamma \in \{1, 10\}$, aspect 1 and 4, single particle and three-particle cloud [worker, 2026l].

Table 6 confirms the reference-separated picture independently. $\gamma = 1$ crosses 3×10^{-4} at $d = 12$ (aspect 1, 1.1×10^{-4}) and $d = 32$ (aspect 4, 2.6×10^{-4}); at $\gamma = 10$, aspect 1 the relative error is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$, so it does not cross 3×10^{-4} within 48 cells. These are floored electric values; the unfloored magnetic error at $\gamma = 10$ is 9.7×10^{-4} at $d = 32$. The window starts at $d = 2$, so the TSC-support interior ($|s| < 1.5$ cells) is not covered by these sweeps; it is what the near shell measures (Sections 3.7, 6.7) [worker, 2026l,m,h].

Fig. 2 plots the mesh against the TSC-convolved continuum; the point-field column of Table 5 (M–P) is the mesh against the un-smoothed field, which carries the TSC smoothing in addition to the deposit error and never agrees with $M-T$ for $\gamma h \gg 1$.

6.4 Crossover radii

At $\gamma = 1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d = 8$; the two are interchangeable there. From $\gamma = 10$ up, against the point field the perpendicular direction does not reach 3×10^{-4} within 48 cells; against the target the perpendicular crossover is 48 ($\gamma = 10$, aspect 1) or not reached. The axial direction stays at 6 against the target at every γ [worker, 2026d].

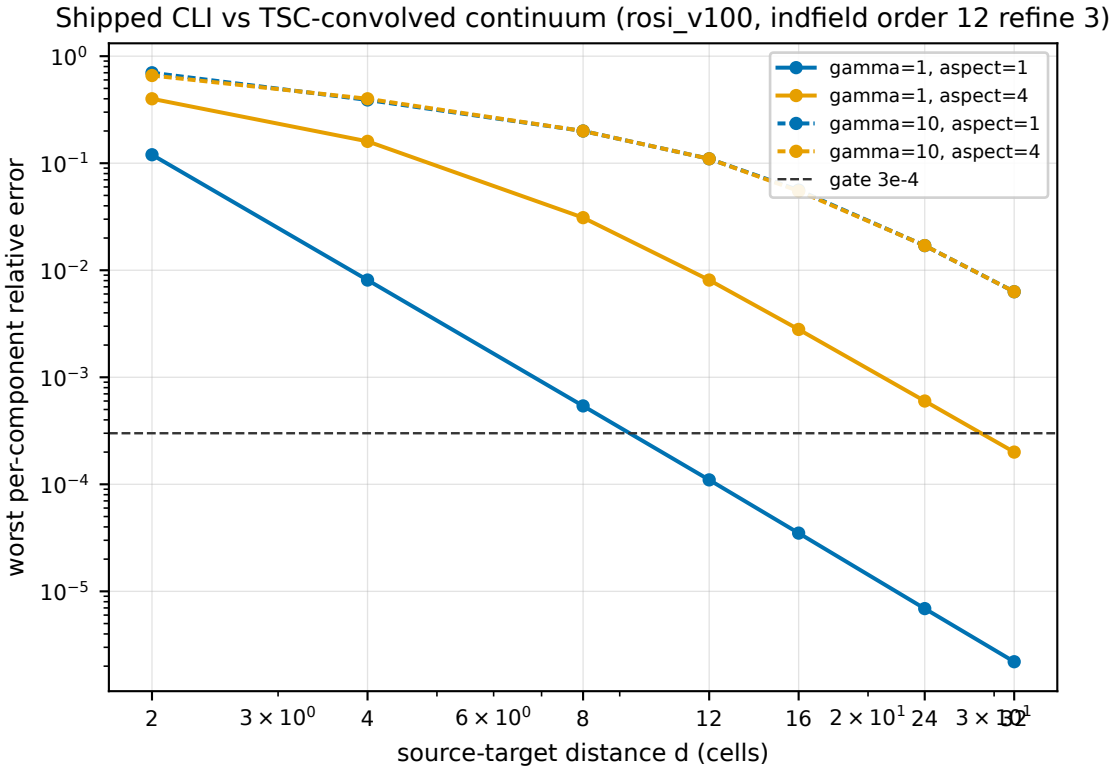


Figure 2: Worst per-component relative error of the shipped CLI against the TSC-convolved continuum, per d in cells, single-particle series (the three-particle cloud is in Table 6); rosi_v100, `indfield.py` order 12 refine 3 [worker, 2026l]. Solid, $\gamma = 1$; dashed, $\gamma = 10$; blue, aspect 1; orange, aspect 4. $\gamma = 1$ crosses the $3{\cdot}10^{-4}$ gate at $d = 12$ (aspect 1) and $d = 32$ (aspect 4); $\gamma = 10$ is $6.5{\cdot}10^{-3}$ at $d = 32$, $1.8{\cdot}10^{-3}$ at $d = 48$. Floored electric series; the magnetic field is excluded by the floor.

6.5 Mid-bunch slices: field and error side by side

The distance sweeps above sample the far field. To see where the single-mesh implementation actually fails, we slice the mesh through the bunch core and compare, pixel by pixel, with the independent exact TSC driver [worker, 2026n,o,p]. The case is $N = 1000$ in a $32 \times 32 \times 16$ box, cell 2×10^{-7} m, rosi_v100: a $\gamma = 10$ bunch (axial, $u_x = 9.95$) and the stationary $\gamma = 1$ bunch. Each figure draws the x-y plane at $z = 1.60\,\mu$m (top) and the z-y plane at $x = 3.20\,\mu$m (bottom), both through the bunch centre; the left column is the mesh field, the right column the relative error, pixels below 10^{-3} of the plane peak masked white. Fig. 3 is the $\gamma = 10$ case (five components; B_x identically zero for an axial current) and Fig. 4 the stationary bunch (B zero, three electric components).

Shell radius unit. These slices run the default mesh with no near shell. The near-shell envelope measured in the crossover work, "3 to 4 PIC cells," assumed a mesh spacing $h = 1/4$ of the PIC cell, so $R = 3$ and $R = 4$ are 12 and 16 mesh nodes [worker, 2026f,p]. On this grid the node spacing is $h = 1$ PIC cell (2×10^{-7} m), so the same $R = 3/4$ is $3/4$ mesh nodes, four times too small in physical units. The slices therefore show a shell configuration that is too small to cover the near field, and that is why they do not reach 10^{-3} (Section 6.6).

Table 8 lists the measured worst relative error per component and the fraction of unmasked plane pixels above 10^{-3}.

The error is largest exactly where the single-mesh implementation has no near-field treatment: these planes cut the bunch core, the region the spectral far-field convolution does not resolve, and the maps show the spatial structure of that failure, concentrated at the beamed core and falling away toward the plane edges. The far field is measured separately (Table 6, Fig. 2); the near-shell requirement is motivated by the core, not by the far field.

Table 7: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: not reached) [worker, 2026d]. T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger. The $\gamma = 1$ aspect 1 row is $d = 8$ against both references here, versus the $d = 12$ of the shipped-CLI series in Table 6: the two use different component sets (six axis directions, L2-pooled, versus per-component over x, y) and different implementations [worker, 2026d,l].

	3×10^{-4}		10^{-3}			
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

6.6 Near-shell accuracy and cost on the slice geometry

On the same $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m geometry, the exact near shell was run through the library path at the radius the envelope case assumed, $R = 3$ for $\gamma = 1$ and $R = 4$ for $\gamma = 10$, in mesh nodes of this $h = 1$ grid [worker, 2026p]. The mesh-only plane errors (Table 8, Fig. 3, Fig. 4) are 0.12–0.41 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$: the single-mesh limitation inside the bunch core, where the planes cut the sources. With the near shell the worst per-component plane error drops to 1.6×10^{-3}–1.1×10^{-2} for $\gamma = 1$ and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$: at $R = 3/4$ on an $h = 1$ grid the shell still does not cover the whole TSC support (Section 6.5), so the residual is again mesh-only. The discrete Gauss-law indicator, the Yee backward-difference $\operatorname{div} E$ residual normalised to the peak deposited ρ/ε_0, moves from 0.3534 to 0.0895 for $\gamma = 1$ and from 5.026 to 0.399 for $\gamma = 10$, against reference values 0.0904 and 0.0814: $\gamma = 1$ reaches the reference value, $\gamma = 10$ drops 12.6× and stays 4.9× above it [worker, 2026p,m].

The assembly itself is not the limiting term: the tests of Section 3.7 show the tiled subtraction exact to roundoff, so the residuals above are the mesh outside the shell [worker, 2026h].

Runtime at $N = 1000$ is 4.3 min for $\gamma = 1$ and 59 min for $\gamma = 10$ (one 16-thread socket) [worker, 2026p,f]; the exact evaluator costs about 8× more per pair at $\gamma = 10$ than at $\gamma = 1$.

6.7 The near-field cost ladder

Measured per-pair costs on `rosi_v100`, single thread [worker, 2026j]: the ported box8/rem8 stationary evaluator (4480 nodes per pair) is 5.4×10^{-5} s, accurate to about 10^{-3} in support only; the accurate moving evaluator is 3.1×10^{-4} s out of support and 6.9×10^{-3} s in support; the upstream adaptive settings (1.07×10^9 nodes) are 1.62 s. The accurate moving path is the one that covers moving sources (Section 3.9). At 2×10^6 particles the in-support pair count is 1.7×10^8, and no measured CPU configuration reaches 2×10^6 particles, 10^{-3} error, and 300 s.

[TODO — UNFINISHED MEASUREMENT: the final 300 s verdict is still being measured. No result is reported here. Insert the result at this marker with its host, particle count, radius, evaluator configuration, and measured wall time.] The statement above holds with or without that measurement: the measured per-pair ladder does not reach the 2×10^6 / 10^{-3} / 300 s envelope on the CPU path, and the placeholder reports the GPU path when it lands.

6.8 The disagreement is per-source

On a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$), the disagreement with the exact-quadrature reference does not grow with particle count: outside the TSC support the worst per-component error

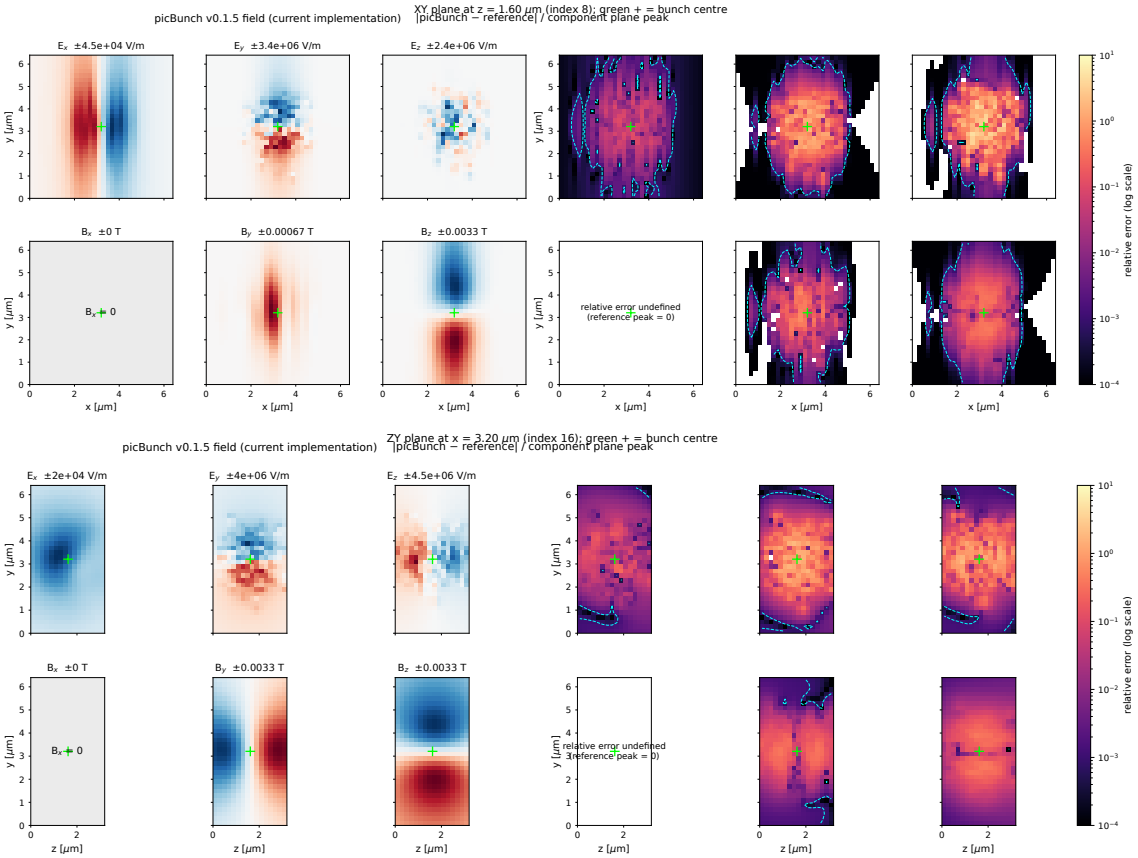


Figure 3: Mid-bunch slices, $\gamma = 10$, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m [worker, 2026n]. Top: x-y plane at $z = 1.60\,\mu$m; bottom: z-y plane at $x = 3.20\,\mu$m. Left column of each panel: mesh field per component; right column: relative error against the independent exact TSC driver (log scale), pixels below 10^{-3} of the plane peak masked white. B_x is zero for an axial current. Mesh-only, no near shell: the panels show the single-mesh limitation inside the bunch core (Section 6.5).

stays in the 10^{-3} to 3×10^{-2} band across $N = 1$ to 1000 [worker, 2026q]. The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.42–0.65 cells from a source, inside that source's TSC support. A flat per-source floor is what a kernel or reference-type difference produces; a charge-weighting defect would grow with N [worker, 2026q].

6.9 Anisotropic cells: the gap is the TSC convolution

At anisotropic spacing the accuracy is quantified per geometry, not inherited from the cubic case [worker, 2026r,s]. On a grid with per-axis spacing $(1, 1, 4)$ m the dominant gap at the near observation points is the TSC convolution itself: the TSC-convolved continuum is 0.9638× the point Coulomb field, a 3.2% offset the mesh reproduces. At 12 cells the mesh matches the continuum to 0.44% at that geometry. The mesh is not wrong because the spacings differ; the 3.2% is the shape the deposit applies. Along the stretched axis the field agrees with the continuum to the same 10^{-3} scale as the cubic control; along the short axes the near points sit in the TSC regime at a short physical distance, a resolution limit, not an anisotropy defect [worker, 2026r,s].

The h^3 factor. The sampled convolution carries the Riemann volume of the continuum integral of Eq. (1): each sample covers h^3 cells, so the read-out applies $h_x h_y h_z/(M_x M_y M_z)$ (Section 3) [extraction worker, 2026].

6.10 Determinism and the `gridUnitSI` defect

Repeated runs of the shipped CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes) [worker, 2026l]. The CLI also rejects a below-

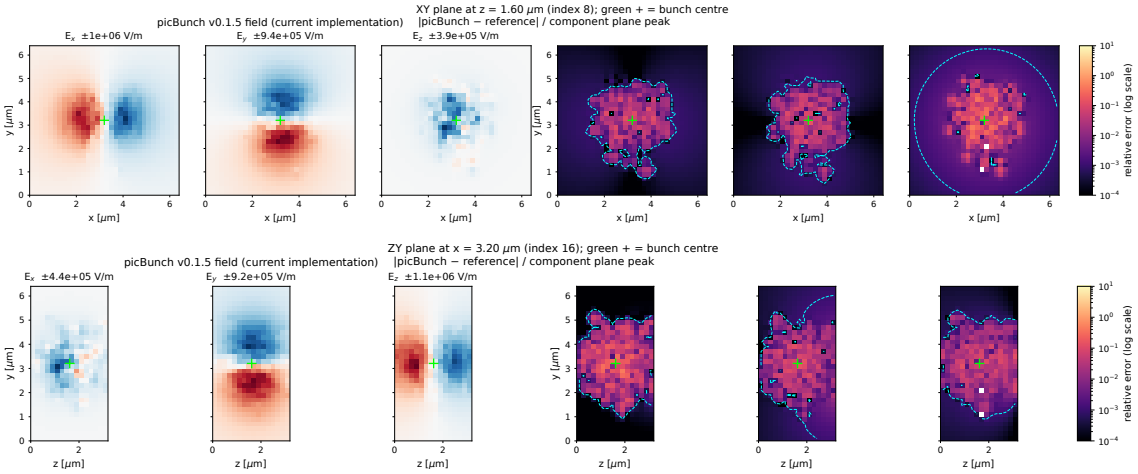


Figure 4: As Fig. 3, stationary bunch ($\gamma = 1$), $N = 1000$, same geometry [worker, 2026o]. B is zero, so only the three electric components are drawn. Mesh-only, no near shell: the single-mesh limitation inside the bunch core.

floor `-tolerance` (10^{-15}) and a two-velocity input, each with exit 1 [worker, 2026l].

The reader and the CLI apply `gridUnitSI` exactly once, in `OpenPMD.cpp:505` and `main.cpp:248` respectively, so a mesh with `gridUnitSI` $= 10^{-6}$ and a physical cell of 10^{-6} m keeps that cell rather than producing 10^{-12} m. An end-to-end test covers this case [worker, 2026l].

7 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Neither is a truth comparison.

7.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 6), evaluated in PIConGPU in its reference-only field-initializer mode [worker, 2026q]. picBunch is at `339ca7ad` (`experimental/mvp`); `ctest` passes 10/10 [worker, 2026t]. Both sides use identical Yee offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation [worker, 2026t].

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B) [worker, 2026t, Orchestrator, 2026d]. picBunch is about four to six orders of magnitude above that floor, a no-agreement statement at the reference floor, not a defect: B is exactly zero for $\gamma = 1$ in both, the bulk agrees to below 1–3%, and the error is concentrated at the core and the beamed peaks [worker, 2026t, Orchestrator, 2026d].

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8×10^{-3} with an L2 of 4.6–5.7×10^{-2}; $\gamma = 10$ bulk median 1.4–3.3×10^{-2} but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92 [worker, 2026t]. The forward E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake that TSC and the spectral near-field kernel cannot resolve [worker, 2026t].

Caveat: the sampling groups are distance-from-centroid. The 318-point groups (inside-support, face, outside) were cut by distance from the bunch centroid, in a bunch with a $\sim$ 6-cell spread [Orchestrator, 2026d]. That grouping may understate how close the "outside" points actually are to a source's TSC support, so the per-group numbers are a spatial profile, not a clean inside/outside-of-support split. The per-source dependence in Section 6 uses the explicit support union and is the measurement that separates the per-source floor from any superposition contribution [worker, 2026q].

Table 8: Mid-bunch slice statistics, mesh-only. Worst relative error per component and fraction of unmasked plane pixels above 10^{-3}, x-y plane at $z = 1.60\,\mu\text{m}$ and z-y plane at $x = 3.20\,\mu\text{m}$, $N = 1000$, $32 \times 32 \times 16$, cell $2 \times 10^{-7}\,\text{m}$, mesh against the independent exact TSC driver (branch `experimental/neuralbunch-hang`, HEAD `e79ddb389`, tolerance 10^{-4}) [worker, 2026n,o]. Node spacing $h = 1$ PIC cell; no near shell. Driver's own floor: $\gamma = 10$ max error 1.16×10^{-3} (4664 of 9.8×10^7 pairs not converged); $\gamma = 1$, 9.075×10^{-4} (14).

plane	component	worst rel. err	frac $> 10^{-3}$
$\gamma = 10$			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.12×10^{-1}	48.6%
	E_y	1.59	51.0%
	E_z	4.60	65.5%
	B_y	6.20×10^{-1}	60.6%
	B_z	3.98×10^{-1}	52.9%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	2.84×10^{-1}	93.6%
	E_y	1.83	90.2%
	E_z	1.93	92.4%
	B_y	3.63×10^{-1}	90.0%
	B_z	4.07×10^{-1}	99.8%
$\gamma = 1$ *(stationary)*			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.20×10^{-1}	22.8%
	E_y	1.34×10^{-1}	23.3%
	E_z	4.15×10^{-1}	74.7%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	4.15×10^{-1}	49.6%
	E_y	1.37×10^{-1}	61.7%
	E_z	1.23×10^{-1}	53.9%

7.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0, E, B $x/y/z$ at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata [worker, 2026u]. The default precision is uniform `float32` for every field and particle record, so a stock single-precision PIConGPU can load the file [worker, 2026b]. `v0.5.0` copies every species from the input bunch and carries a single checkpoint-wide `picongpu_idProvider` range [worker, 2026v,g,a]; the `charge` and `mass` records are written rather than omitted, and the species is a single particle patch (Section 3.1). The remaining differences against a genuine PIConGPU checkpoint are non-blocking metadata [worker, 2026u].

Field arrival is bitwise. A fresh `v0.5.0` checkpoint written from a two-species input (16 electrons and 16 ions, `picongpu_idProvider` startId 0, nextId 32) restarts in PIConGPU with exit 0 under both `patch` modes (`bbox`: offset $(7, 8, 3)$, extent $(8, 10, 6)$; `domain`: offset $(0, 0, 0)$, extent $(24, 24, 12)$), all six field components bitwise-equal, $\max_{\text{abs diff}} = 0.0$, electrons with ids 0 to 15, first-step displacement exactly $\beta c\,dt$ with $\beta = 0.9949874367$ [worker, 2026a]. A validation-only build with the field solver set to `None` loads the checkpoint and writes step 1; all six components are bitwise-equal, max ULP 0 [worker, 2026u].

7.3 PML passthrough and the restart fixes

On a $24 \times 24 \times 12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the $\sim 116\,\text{s}$ neuralBunch path to 1.8 s because the checkpoint carries the field [worker, 2026u]. The same investigation found the defects in PIConGPU's restart path that the fixes PR addresses [Orchestrator, 2026e]: `RestartFieldLoader` never reads `unitSI`, `gridSpacing` or `gridGlobalOffset`; the PML slabs are unconditional; a successful restart suppresses the initial dump; and the openPMD backend does not convert dataset dtypes. The fixes PR (`psychocoderHPC-bot/picongpu` PR #25, draft) adds restart metadata validation, unitSI conversion, optional PML slabs, and an explicit dtype contract, verified by real restarts including a factor-2 unitSI case [Orchestrator, 2026e].

Table 9: Whole-field comparison of picBunch against the PIConGPU `referenceOnly` field, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100 [worker, 2026t]. `worst_rel` is peak-normalized; median and p99 are per-point over nodes with $|\text{ref}| > 10^{-3} \times$peak; L2 is pooled. Not a truth comparison.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

8 Discussion and limitations

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method.

What it is good for. In the far field the relative error against the point field is below 10^{-3} on the beam axis from $d = 6$ for every γ [worker, 2026d]. Against the TSC-convolved continuum: $\gamma = 1$, aspect 1 crosses 3×10^{-4} at $d = 12$ (1.1×10^{-4}); $\gamma = 1$, aspect 4 at $d = 32$ (2.6×10^{-4}); $\gamma = 10$ is 6.5×10^{-3} at $d = 32$, not below 3×10^{-4} within 48 cells [worker, 2026l,m]. At 2×10^6 particles in a 20-cell box the compute-only total is 64.8 ms, dominated by the deposit (56.5 ms), and flat in γ [worker, 2026c]. Repeated runs are bit-identical (md5 `d3641d64...`); the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14} [worker, 2026l,k].

What it is not. The mesh alone does not reach 10^{-3} where the slices cut the bunch core: on the $N = 1000$ slice geometry the mesh-only worst per-component plane error is 0.12–0.41 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ (Table 8). The exact near shell (Section 3.7) is opt-in and its assembly is exact to roundoff; at the measured $R = 3/4$ radii on the $h = 1$ slice grid the residual is the mesh outside the shell, not the tile subtraction (Section 6.6). The perpendicular direction at $\gamma = 10$ does not converge at the natural cell size (1.5×10^{-2} at $d = 48$, aspect 1; the TSC smoothing alone is 1.48×10^{-2}, Table 5), and reaching 10^{-3} perpendicular at $r = 8$ cells would need $h \approx 1/29$, a 2.4×10^4-fold increase in M^3 FFT work [worker, 2026d, Orchestrator, 2026b]. At aspect 4, $\gamma = 1$ already fails the gate at $d = 48$ (2.4×10^{-3}) [worker, 2026d]. The cross-validation confirms the same boundary: the mesh does not agree with the exact reference at the reference floor; the disagreement is per-source, in the TSC support, and does not grow with particle count [worker, 2026q, Orchestrator, 2026d].

Performance. The cost is two regimes with a crossover set by the box, not the particle count [worker, 2026c]. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows. At large boxes the FFT and kernel dominate: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× [worker, 2026c]. The compute timings are single-GPU on one `rosi_v100` host. The output path has a multi-rank measurement: at 512^3, tolerance 10^{-3}, the chunked writer runs in 54.3 s for one rank and 43.2 s for two, the six datasets bitwise identical (Section 4.1) [worker, 2026g]. The near-field cost is a separate wall: per-pair, 5.4×10^{-5} s for the ported stationary box8/rem8 evaluator, 3.1×10^{-4} s out of support and 6.9×10^{-3} s in support for the accurate moving evaluator, 1.62 s for the upstream adaptive settings (Section 6.7) [worker, 2026j].

Scope. The defensible scope, in floored relative per-component error against the continuum, is $\gamma = 1$ aspect 1 from $d = 12$, $\gamma = 1$ aspect 4 from $d = 32$, and the axial component at $\gamma \leq 10$ [worker, 2026l,d, Orchestrator, 2026b]. The blanket statement "$\gamma \leq 10$ meets 10^{-3}" is not supported: it fails perpendicular at the natural cell size, 6.5×10^{-3} at $\gamma = 10$, $d = 32$, aspect 1, still 1.8×10^{-3} at $d = 48$, not below 3×10^{-4} within 48 cells [worker, 2026l,d,m]. Each statement in Section 6 names which metric (floored,

unfloored, absolute) supports it; the floored column is electric-only at $\gamma = 10$ because the floor excludes the magnetic field, and the judge metric $|\text{ref} - \text{mesh}|/(1 + \max)$ reports $d = 2$ for every case and does not discriminate. The radii in Table 7 are axis directions only; oblique directions are $\sim 1.5\times$ larger. Scale invariance (Section 3.6) means only r/h moves the error, so reaching 10^{-3} at $r = 8$ needs $h \approx 1/29$ (2.4×10^4 times the M^3 nodes) [worker, 2026d, Orchestrator, 2026a].

9 Conclusion

`picBunch v0.5.0` computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, driven by a single TOML configuration (Section 3.1), and writes a uniform float32 checkpoint that restarts under a stock PIConGPU. In floored relative per-component error against its TSC-convolved target, the mesh is below 3×10^{-4} from $d = 12$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$ it is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$, not below 3×10^{-4} within 48 cells [worker, 2026l,m]. The disagreement with the PIConGPU reference field is per-source, local to the TSC support, and independent of particle count. The mesh alone does not reach 10^{-3} where the slices cut the bunch core (0.12–0.41 for $\gamma = 1$, 0.11–4.60 for $\gamma = 10$); the opt-in near shell (Section 3.7) assembles exactly to roundoff, so the residual at the measured radii is the mesh outside the shell, not the tile subtraction [worker, 2026h]. The near-field cost ladder is 5.4×10^{-5} s per pair for the ported stationary box8/rem8 evaluator, 3.1×10^{-4} s out of support and 6.9×10^{-3} s in support for the accurate moving evaluator, 1.62 s for the upstream adaptive settings (Section 6.7); the MPI chunked output is bitwise identical across rank counts and the $6R$ far-tail switch is 1.87×10^{-3} of the target [worker, 2026g]. On a two-species periodic deck the field round-trips bitwise through a real PIConGPU restart, first-step displacement exactly $\beta c\,dt$ with $\beta = 0.9949874367$ (Section 7). The method is a far-field tool: cheap, deterministic (md5 `d3641d64...`), and GPU-portable, with the mesh limitation inside the bunch core documented by the slices and the perpendicular direction at $\gamma \geq 10$ not converged at the natural cell size.

References

S1 extraction worker. picbunch portable algorithm specification (s1), 2026. Run pr24-20261102. S1-algorithm-spec.md. Exact field definition and discrete recipe with path:line provenance. meshconv.cu on branch fee89ed75; ExactShapeField.hpp on 2baa2566.

INT2 worker. picbunch v0.5.0 release: placement interface (int2), 2026a. Run pr24-20261102. INT2/RESULTS.md. v0.5.0, commit 7b4f1766, tag 3e083fef. TOML interface, charge-weighted centring, float32 default, charge/mass records, patch = bbox or domain. Restart: bbox and domain exit 0, six components bitwise, ids 0-15, idProvider 0/32, first step beta*c*dt. ctest 18/18.

IMPL3 worker. Uniform checkpoint precision and stock-picongpu restart (impl3), 2026b. Run pr24-20261102. IMPL3/RESULTS.md. precision = float32 default for every field and particle record; float64 uniform opt-in with a picBunchPrecisionWarning attribute. float32 restart on rosi_v100: exit 0, six components bitwise, ids 0-15, first step beta*c*dt. Branch experimental/placement, commit b68a141bb.

SB1 worker. Measured scaling of the picbunch field pipeline (sb1), 2026c. Run pr24-20261102. SB1/RESULTS.md. V100 timing matrix, Coulomb limit, distance sweep. Commit dc69c16a.

N2b worker. picbunch mesh vs the tsc-convolved continuum and the point field (n2b), 2026d. Run pr24-20261102. N2b/RESULTS.md. Reference-separated accuracy, scale invariance, crossover table. Arbiter: indfield.py.

N1b worker. picbunch mesh convolution measurements, cpu (n1b), 2026e. Run pr24-20261102. N1b/RESULTS.md. Wraparound control, R2C/C2R vs C2C, padding vs distance, kernel sampling rule, CPU cost model. CPU only, no GPU.

Orchestrator. Decision d10: mesh accuracy is scale-invariant in h at fixed cell distance, 2026a. Run pr24-20261102. DECISIONS.md, D10. Kernel homogeneous of degree −2, TSC width equals node spacing; identical errors at h = 1, 1/2, 1/4, 1/8.

NS2 worker. picbunch exact per-pair near shell (ns2), 2026f. Run pr24-20261102. NS2/RESULTS.md. F = F_mesh(all) - F_mesh(near) + F_exact(near), opt-in –near-shell exact, per-tile local FFT, vendored FP64 TSC reference. Envelope and cost (3.1e-4 s per pair) in the text. Commit 9733d232.

FIX1 worker. picbunch v0.4.1 release: near-shell cli and far-tail margin (fix1), 2026g. Run pr24-20261102. FIX1/RESULTS.md, margin_table.txt. v0.4.1, commit 02d2d06e, tag 9e4fd7dc. Near-shell CLI parsed; quadrupole far-tail bound and 6R switch (switch error, 512-cubed MPI equality in the text). Checkpoint: all species, checkpoint-wide idProvider, restart bitwise. ctest 16/16 (rosi sm_70, jureca-h100 sm_90, hal gfx1100 HIP).

NS4b worker. Near-shell assembly diagnosis: subtraction is exact (ns4b), 2026h. Run pr24-20261102. NS4b/RESULTS.md. Tile-centre/boundary residuals 2e-12 to 1.5e-11 SI (4e-16 rel); local vs global FFT 1e-15; 100-particle reproduction: assembled 4.6e-13 vs plain mesh 5.6e-1. The 1.36e-1 plane error is mesh-outside-the-shell, not assembly. v0.4.1 (02d2d06), rosi_v100.

M1 worker. picbunch global-domain output and mpi (m1), 2026i. Run pr24-20261102. M1/RESULTS.md. –global-grid/–global-origin, chunked writer, PICBUNCH_ENABLE_MPI. 512 cubed bitwise equality across rank counts; rosi_v100, OpenMPI 4.1.6, openPMD 0.17.1.

K3 worker. In-support evaluator port and per-pair cost ladder (k3), 2026j. Run pr24-20261102. K3/RESULTS.md. O(1) asinh box verified against a converged trapezoid reference (-2.6333 vs -2.6348); octant-split remainder; power-transform quadrature the validated moving path (1.000000/1.000000/1.000001/1.000040 at gamma 1/2/10/100, 1.5e-6 at box order 32); no validated closed form for the moving case. Per-pair costs, rosi_v100 single thread: ported box8/rem8 5.4e-5 s; accurate moving 3.1e-4 s out of / 6.9e-3 s in support; upstream adaptive (1.07e9 nodes) 1.62 s.

pr24w53 ref. indfield.py: independent tsc-convolved continuum reference, 2026. sha256 a1355980. Tensor Gauss-Legendre with TSC-knot split and a singularity-cancelling power map; the independent truth for the mesh.

SB1 worker. Coulomb-limit control (sb1), 2026k. SB1/RESULTS.md and PF/data/coulomb.csv. u = 0, one particle at a node, far nodes at d = 8, 16, 24, 32, six axis directions. Worst relative error 4.88e-14 (max absolute 2.28e-25); threshold 1e-12: PASS.

W4b worker. Independent validation of the shipped picbunch (w4/w4b), 2026l. Run pr24-20261102. W4b/RESULTS.md. picBunch at bfaa1e26 (experimental/mvp), rosi_v100, ctest 8/8. Shipped CLI vs indfield.py order 12 refine 3, relative per-component error per d = 2 to 32 cells, gamma 1 and 10, aspect 1 and 4.

Orchestrator. Decision d11: v0.1 accuracy scope, 2026b. Run pr24-20261102. DECISIONS.md, D11. Not defensible: gamma <= 10 meets 1e-3. Defensible: gamma <= 1 aspect 1; gamma <= 10 axial only; gamma 100 axial only.

Orchestrator. Decision d8: gamma-100 outside-band 9.43e-2 is unresolved, 2026c. Run pr24-20261102. DECISIONS.md, D8. Pre-existing 9.43e-2 at 1.6 to 6 cells, gamma 100 axial, outside the near box.

V2b worker. No-floor accuracy audit of the shipped picbunch executable (v2b), 2026m. Run pr24-20261102. V2b/RESULTS.md. Three-number metric (floored, unfloored, absolute) per case and distance; gamma10 B excluded by the floor (B 1e-10 of the local field), unfloored B 9.683e-4 at d=32. picBunch main v0.3.0 a12ce3fa.

SL1 worker. Mid-bunch 2-d slice figures, gamma 10 (sl1), 2026n. Run pr24-20261102. SL1/RESULTS.md, data/slices_g10.npz. picBunch v0.1.5 (5f1981e2) vs the independent exact TSC driver (HEAD e79ddb389, tol 1e-4). N=1000, 32x32x16, cell 2e-7 m, rosi_v100; planes x-y at z=1.60 um, z-y at x=3.20 um.

SL2 worker. Mid-bunch 2-d slice figures, stationary gamma 1 (sl2), 2026o. Run pr24-20261102. SL2/RESULTS.md, data/slices_g1.npz. Same geometry and reference as SL1 (driver max error 9.075e-4, 14 not converged). B identically zero; electric components only.

SL3 worker. Mid-bunch slices and near-shell acceptance (sl3), 2026p. Run pr24-20261102. SL3/RESULTS.md. N=1000, 32x32x16, cell 2e-7 m, rosi_v100, v0.4.0 (bc0037c5). Default-mesh and near-shell (R=3/4, mesh nodes, h=1) plane errors, Gauss-law residual, runtime; the 3/4 envelope assumed h=1/4, so at h=1 the shell is 4x too small. Driver HEAD e79ddb389.

W7t worker. Reference-quantity identification and particle-count scaling (w7t), 2026q. Run pr24-20261102. W7t/RESULTS.md. The reference is the TSC-convolved field, the same quantity picBunch computes. N-count scaling: disagreement is per-source, not superposition; B identically zero for the stationary bunch.

A1b worker. picbunch anisotropic-cell correctness, per-axis sampling (a1b), 2026r. Run pr24-20261102. A1b/RESULTS.md. Per-axis sampling is correct (FFT equals the direct discrete TSC sum to 4e-16 on cell 1,1,4); the anisotropic gap is the TSC convolution itself (continuum 0.9638 times point Coulomb, 3.2 percent; mesh matches continuum to 0.44 percent at 12 cells). picBunch 339ca7ad.

A1c worker. picbunch anisotropic-cell field correctness, cli (a1c), 2026s. Run pr24-20261102. A1c/RESULTS.md. v0.1.3 da3cd42. The stretched axis (dz=4 m) of a (1,1,4) grid matches indfield.py to about 1e-3, the cubic-control scale; the short x/y axes sit in the near-field TSC regime. Per-geometry quantification, not an anisotropy defect.

W7r worker. picbunch cross-validation against the picongpu field reference (w7r), 2026t. Run pr24-20261102. W7r/RESULTS.md. picBunch @ 339ca7ad. 318 staggered points (64x64x32, cell 1e-7 m) and whole-field (N=1000, 32x32x16, cell 2e-7 m); groups distance-from-centroid (caveat D13).

Orchestrator. Decision d13: cross-validation does not show agreement at the reference floor, 2026d. Run pr24-20261102. DECISIONS.md, D13. 3.66e-2 (gamma 1), 5.16e-1 (gamma 10 E/y) vs floor 1.19e-6 (E), 1.43e-6 (B); no agreement claim with PIConGPU.

W7b worker. picbunch checkpoint acceptance on picongpu pr 24 (w7b), 2026u. Run pr24-20261102. W7B/RESULTS.md. Early acceptance state: 16-particle periodic deck restarts exit 0, six components bitwise (0 ulp) via a solver=None load; init 1.8 s vs 116 s. Fix commit 0c9427dee82e01c832c42a8358033b701386f37e.

INT1 worker. picbunch v0.4.0 integration release (int1), 2026v. Run pr24-20261102. INT1/RESULTS.md. Integration commit bc0037c5, tag v0.4.0; checkpoint writer copies every species, MPI 512-cubed bitwise, two-species restart bitwise.

Orchestrator. Decision d15: Picongpu fixes pr #25 (restart path), 2026e. Run pr24-20261102. DECISIONS.md, D15. psychocoderHPC-bot/picongpu PR #25 (draft), head bced321c; restart metadata, unitSI, PML slabs, dtype contract.